

Portfolio Management

Rebalancing Portfolios

- Changing portfolio components in a passive portfolio in tune with the benchmark index portfolio is termed as rebalancing portfolios.
- Rebalancing of portfolios is also carried out in actively managed portfolios when there is a requirement of adjusting the components to maintain the initial portfolio characteristics.

Relative Weights of Securities in Portfolios

$$\text{Relative Security weight} = \frac{w_{sp}}{MC_S / MC_B}$$

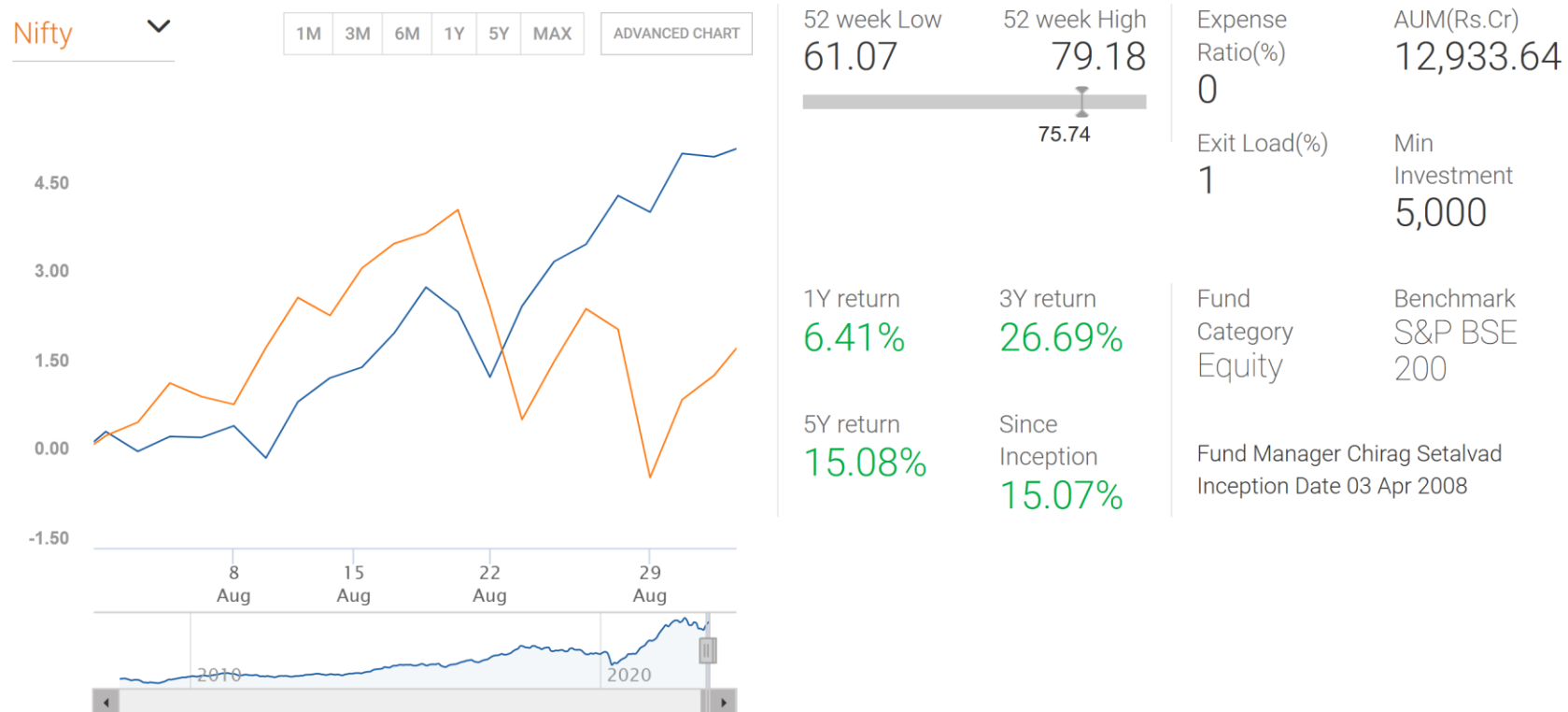
$$\text{Industry Relative weight} = \frac{w_{oi}}{MC_{oi} / MC_B}$$

w_{sp} – Weight of security in a portfolio; w_{oi} – Weight of other industries in a portfolio;

MC_S – Security Market Capitalization; MC_{oi} – Other Industries Market Capitalization;

MC_B – Total Benchmark Market Capitalization.

HDFC Small cap Fund



Source: <https://www.hdfcfund.com>

Portfolio Industry Relative Weight:(0.96 / 1.04)

Benchmark	Weight	HDFC small cap	Weight	Relative weight	Portfolio Revision
Agri	0.35	Agriculture	0.94	0.99	No Revision
Alcohol	0.21	Auto Components	5.82	1.00	No Revision
Automobile & Ancillaries	5.87	Banks	8.05	1.07	Buy
Aviation	0.72	Beverages	2.13	1.05	Buy
Bank	14.01	Capital Markets	0.39		
Capital Goods	1.82	Cash, Cash Equivalents	5.49		
Chemicals	3.7	Chemicals	3.05	1.01	No Revision
Construction Materials	1.91	Commercial Services	4.47	0.99	No Revision
Consumer Durables	0.65	Construction	5.03	0.97	No Revision
Electricals	0.17	Consumer Durables	8.72	0.92	Sell
Finance	7.84	Electrical Equipment	3.62	0.97	No Revision
FMCG	8.4	Entertainment	2.24	0.98	No Revision
Gas Transmission	1.97	Ferrous Metals	2.44	0.99	No Revision
Healthcare	4.44	Fertilizers & Agrochemicals	6.26		
Hospitality	0.45	Finance	2.12	1.06	Buy
Inds. Gases & Fuels	0.55	Healthcare Services	2.93	1.02	No Revision
Infrastructure	1.24	Industrial Products	9.63	0.92	Sell
Insurance	1.98	It - Services	4.18	1.03	No Revision
Iron & Steel	1.66	It - Software	5.08		
IT	12.25	Leisure Services	3.92	0.97	No Revision
Logistics	1.01	Media	0.49	1.00	No Revision
Media & Entertainment	0.2	Personal Products	2.01	1.07	Buy
Textile	0.25	Pharmaceuticals	2.61		
Trading	3.71	Textiles & Apparels	2.16	0.98	No Revision
Miscellaneous	0.11	Transport Services	6.22	0.95	Sell

Constant Proportion Portfolio Insurance (CPPI) Strategy

- Constant proportion portfolio insurance identifies a floor value that is invested in securities.
- A multiplier of excess of portfolio value over floor value is the amount that is retained in the securities and the remaining amount is either kept as cash or portfolio components that exhibit counter cyclical behavior.
- $\text{Value in stocks} = \text{Multiplier} \times (\text{Portfolio value} - \text{Floor value})$

CPPI Strategy

Security	Value (lakh)	% NAV	Estimated Fund Commitment Multiplier	Portfolio Value change	Rebalancing	Revised Portfolio
Bajaj Electricals Ltd.	62,161.52	4.81	4.50	60,296.67	0	60,296.67
Sonata Software Ltd.	57,166.68	4.42	4.50	85,750.02	-25931.6	59,818.45
Firstsource Solutions Ltd.	54,051.55	4.18	4.50	81,077.33	-24506.9	56,570.39
Bank of Baroda	44,540.26	3.44	4.50	66,810.39	-20254.9	46,555.54
eClerx Services	40,470.18	3.13	4.50	32,376.14	0	32,376.14
Aster DM Healthcare	37,864.14	2.93	4.50	34,077.73	0	34,077.73
Gujarat Fluorochemicals	34,530.75	2.67	4.50	34,530.75	0	34,530.75
Chambal Fertilizers & Chemicals	33,825.75	2.62	4.50	37,208.33	-1750.33	35,458
Great Eastern Shipping Company	31,621.93	2.44	4.50	34,784.12	-1762.17	33,021.95
PNC Infratech Ltd.	30,079.55	2.33	4.50	42,111.37	-10578.1	31,533.26
Sharda Cropchem Ltd.	29,171.85	2.26	4.50	29,171.85	0	29,171.85

Multiplier = (Minimum portfolio value)/(Average portfolio value – Minimum Portfolio Value)

Constant Mix Strategy

- Constant mix strategy requires portfolio managers to maintain a relative weight in tune to the benchmark portfolio.
- Constant mix strategy sells stock when its market value increases.

Constant Mix Strategy

Company	Value (lakh)	% NAV	Benchmark Weightage	Relative Security weight
Advanced Enzyme Technologies Ltd.	12,211.78	0.94	0.13	7.23
Apar Industries Limited	9,146.10	0.71	0.38	1.87
Aster DM Healthcare Limited	37,864.14	2.93	1.5	1.95
Aurobindo Pharma Ltd.	5,236.16	0.40	0.14	2.86
Bajaj Consumer Care Ltd.	10,897.55	0.84	0.52	1.62
Bajaj Electricals Ltd.	62,161.52	4.81	0.52	9.25
Bank of Baroda	44,540.26	3.44	0.31	11.10
Carborundum Universal Ltd.	13,447.49	1.04	0.2	5.20
Chalet Hotels Ltd.	15,892.50	1.23	0.2	6.15
Chambal Fertilizers & Chemicals Ltd.	33,825.75	2.62	0.2	13.10
D B Corp Limited	4,171.52	0.32	0.15	2.13
DCB Bank Limited	12,294.66	0.95	0.13	7.31

Constant Proportion Constant Beta Strategy

- Constant beta portfolio rebalances the portfolio by maintaining the initial portfolio beta even after rebalancing.
- Rebalancing is achieved by reducing or increasing the alternate portfolio component such as debt or cash instrument.
- Such reduction or increase is utilized in purchase or sale of equity securities with required beta values.

Constant Beta Strategy

Company	%NAV	Beta	Revised beta		Revised Weight
Advanced Enzyme Technologies Ltd.	0.94	0.54	1.73	Sell	0.92
Apar Industries Limited	0.71	2	0.42	Buy	0.71
Aster DM Healthcare Limited	2.93	0.46	1.4	Sell	2.80
Aurobindo Pharma Ltd.	0.40	0.64	1.89	Sell	0.40
Bajaj Consumer Care Ltd.	0.84	1.83	1.58	Buy	0.83
Bajaj Electricals Ltd.	4.81	1.26	0.67	Buy	4.65
Bank of Baroda	3.44	0.28	1.48	Sell	3.26
Carborundum Universal Ltd.	1.04	0.37	0.26	Buy	1.04
Chalet Hotels Ltd.	1.23	0.26	0.6	Sell	1.22
Chambal Fertilizers & Chemicals	2.62	0.59	1.69	Sell	2.50
D B Corp Limited	0.32	0.94	0.93	Buy	0.32
Portfolio Risk (weighted beta)		90.004	92.8006		90

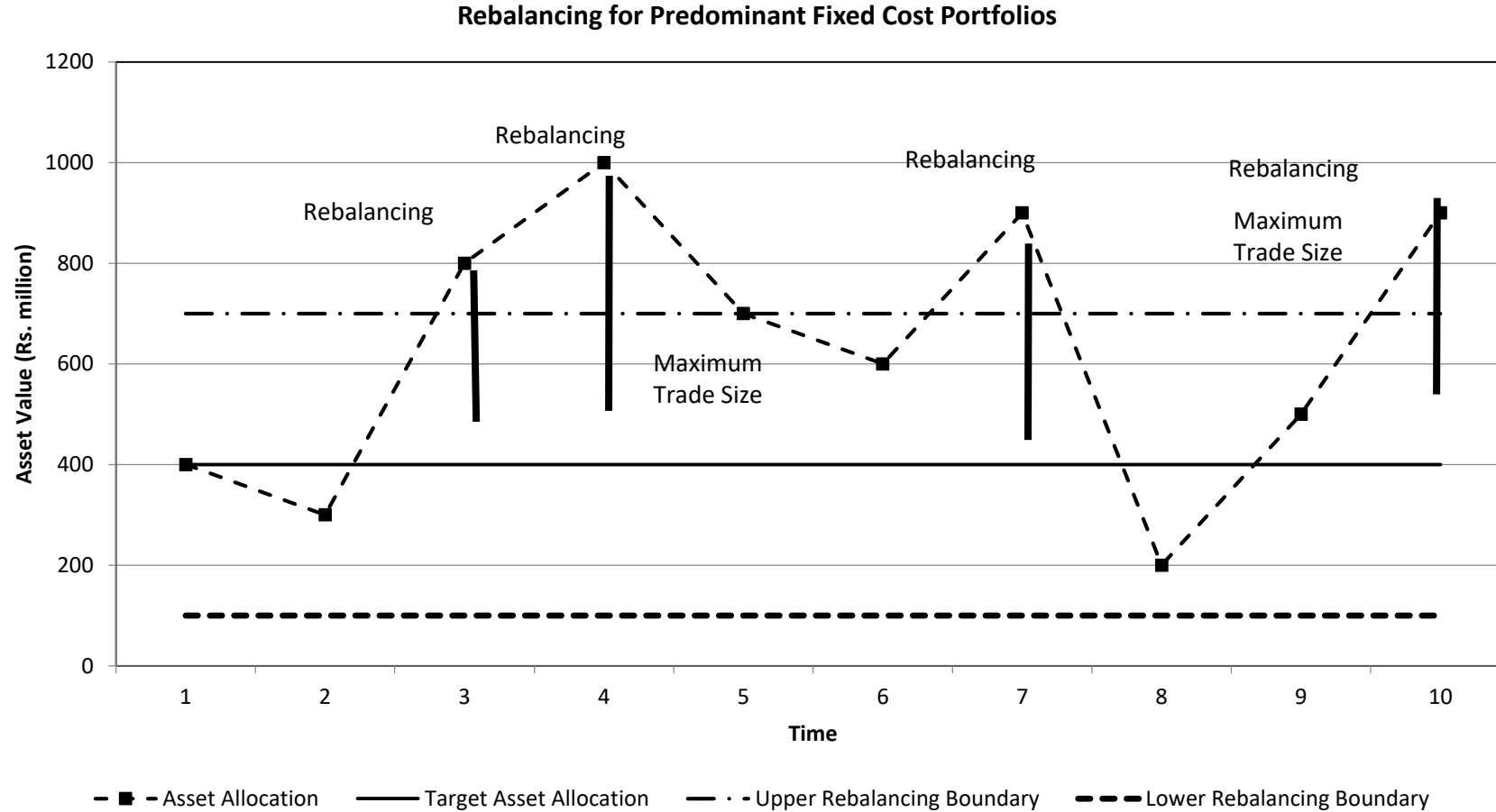
Cost Benefit Analysis of Portfolio Rebalancing

- Transaction costs
- Taxes
- Information costs
- Skill requirement costs

Cost Benefit Analysis of Portfolio Rebalancing

- Fixed costs associated with portfolio rebalancing
- Independent of the trade size
 - Rebalancing skill related payments
 - IT costs
 - Research Analysts, Operations executive, Customer service executive costs
 - Information acquisition costs
 - Management time costs

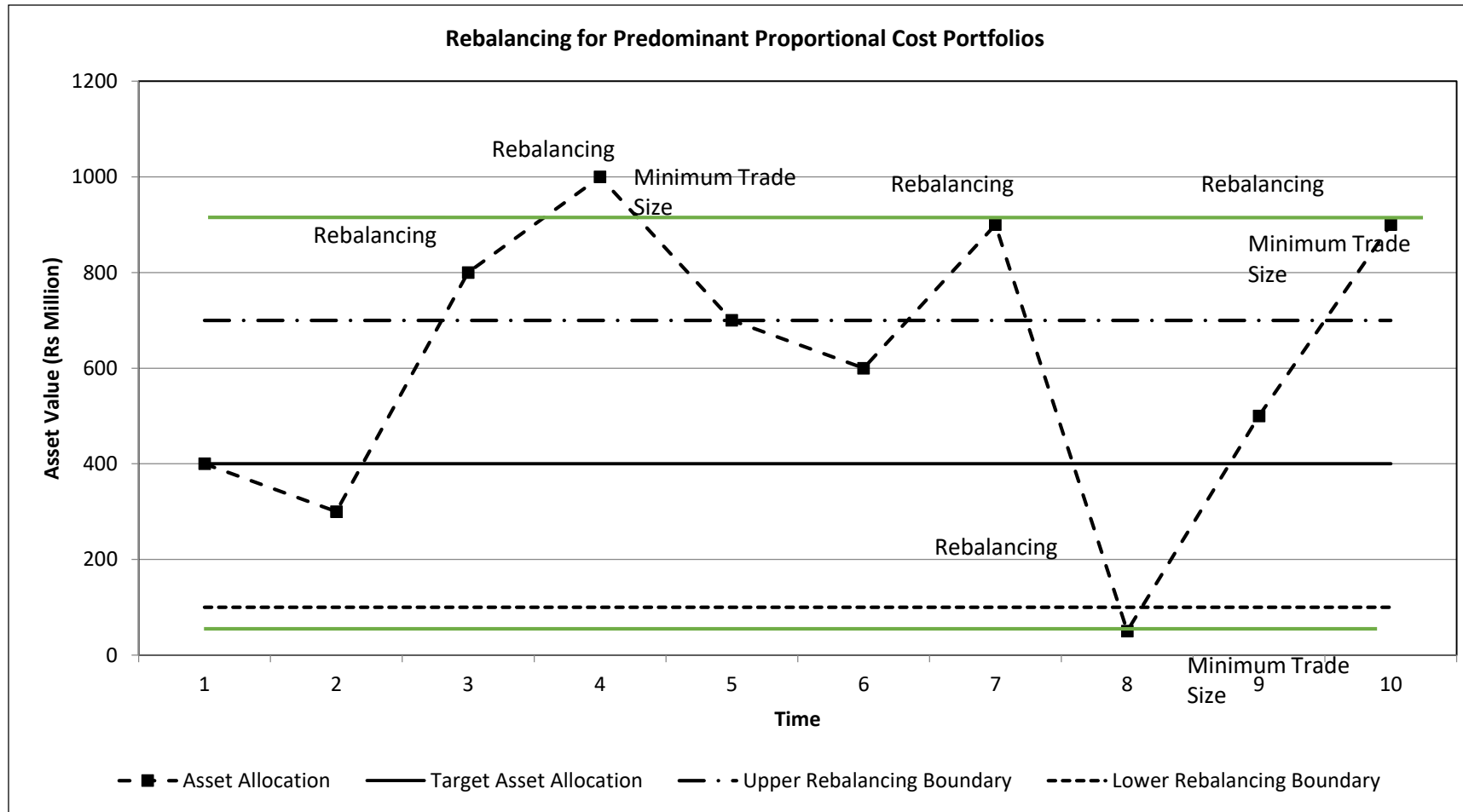
Cost Benefit Analysis of Portfolio Rebalancing



Cost Benefit Analysis of Portfolio Rebalancing

- Variable costs associated with portfolio rebalancing
- Portfolio rebalancing costs can also be proportional to that of order size
 - Redemption fees
 - Taxes
 - Dealer Commission, Trade commission
 - Bid-ask spread
 - Impact cost of trade

Cost Benefit Analysis of Portfolio Rebalancing



Portfolio Revision

- Cost Averaging Plan
- Constant Value Plan
- Constant Ratio Plan
- Variable Ratio Plan

Cost Averaging

Cost Average Formula Plan (amount in INR)							
Month	Market Price	Shares Purchased (No.)	Cumulative Investment	Market Value	Un-realized Profit or Loss	Average Cost Per Share	Average Market Price Per Share
1	100	100	10,000	10,000	0	100	100
2	90	120	20,800	19,800	(1,000)	94.54	95
3	100	100	30,800	32,000	1,200	96.25	96.67
4	110	90	40,700	45,100	4,400	99.27	100
5	120	80	50300	58,800	8,500	102.65	104

Constant Value Plan

Month	Market Price (INR)	Number of Shares	Value of Aggressive (equity) Portfolio (INR)	Value of Defensive (bond) Portfolio (INR)	Total (INR)
1	100	100	10,000	10,000	20,000
2	85	100	8,500	10,000	18,500
3	80	125	10,000	8,000	18,000
Bought 25 shares					
4	85	125	10,625	8,000	18,625
5	95	125	11,875	8,000	19,875
6	97	104	10,088	10,037	20,125
Sold 21 Shares					

Constant Ratio Plan

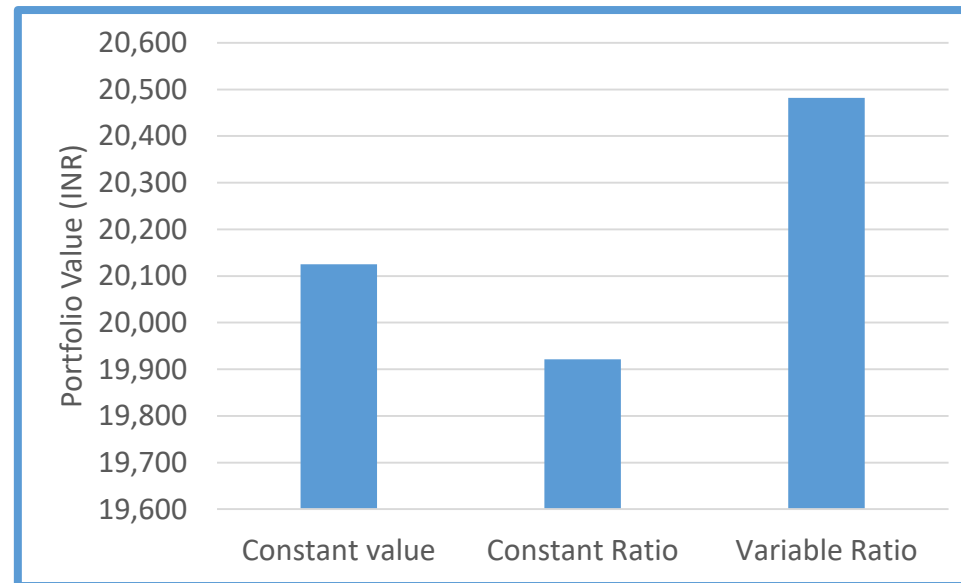
Constant Ratio (1:1) Trigger ratio: (0.80 – 1.2)						
Month	Price (INR)	Quantity	Equity portfolio value (INR)	Debt portfolio value (INR)	Total Portfolio value (INR)	Ratio
1	100	100	10000	10000	20000	1
2	85	100	8500	10000	18500	0.85
3	80	113	9040	8960	18000	1.01
Bought 13 Shares						
4	85	113	9605	8960	18565	1.07
5	95	113	10735	8960	19695	1.2
6	97	103	9991	9930	19921	1.01
Sold 10 Shares						

Variable Ratio Plan

Variable Ratio (Trigger ratios: 0.80 – 2.20) Min Equity 50%; Debt 35%						
Month	Price (INR)	Quantity	Equity portfolio value (INR)	Debt portfolio value (INR)	Total Portfolio value (INR)	Ratio
1	100	100	10000	10000	20000	1
2	85	100	8500	10000	18500	0.85
3	80	100	8000	10000	18000	0.80
3	80	146	11680	6320	18000	1.85
Bought 46 Shares [Hold: (Portfolio value - Portfolio value x 35%) / Price = 146]						
4	85	146	12410	6320	18730	1.96
5	95	146	13870	6320	20190	2.19
6	97	146	14162	6320	20482	2.24
6	97	104	10088	10394	20482	0.97
Sold 42 Shares [Hold: (Portfolio value – Portfolio value x 50%) / Price = 104]						

Comparative Performance

Method	Month	Price (INR)	Quantity	Equity portfolio value (INR)	Debt portfolio value (INR)	Total Portfolio value (INR)	Ratio
Value	6	97	104	10,088	10,037	20,125	1.01
Ratio	6	97	103	9,991	9,930	19,921	1.01
Variable	6	97	104	10,088	10,394	20,482	0.97



Comparison of Portfolio Revision Plans

- Increasing / Decreasing Trend in Prices
 - Ratio plans are able to better utilise the portfolio value in a rising trend
- Volatile markets
 - Constant value plan and variable ratio plans would be competitive portfolio revision tools
- Equally important components
 - Ratio plans perform better
- Components that do not exhibit contra performance relationship
 - Variable plans perform better